

Addressing Modes Exercise

- What are the contents of registers A, R0, R7 and memory locations 30H and 31H after the following code runs: -

MOV A, #5

MOV R7, #40H

MOV R0, #30H

MOV 31H, #14H

MOV @R0, A

INC R0

MOV R7, @R0

- How long does the code take to execute if the 8051 is operating off a 12MHz crystal?

Some Useful Directives

- **END**
 - Last line of code. Assembler will not compile after this line
- **ORG**
 - Origin directive. Sets the location counter address for the following instructions
- **EQU**
 - Equate directive. Used to equate a name with an address or a data value. Useful for constant assignments.
- **DATA**
 - Used to assign a symbol name to an address in internal data memory
AVERAGE DATA 30H
MOV AVERAGE, A
- **BIT**
 - Used to assign a symbol name to a bit in bit-addressable data memory

Programme Sequencing

- Normal program execution is sequential
 - The PC is loaded with the address of instruction N+1 while instruction N is being executed
- The program branching instructions allow the programmer to alter the program execution sequence
 - These instructions allow the address contained in the PC to be changed
 - Program branching is used for jumps, function calls and interrupt service routines.

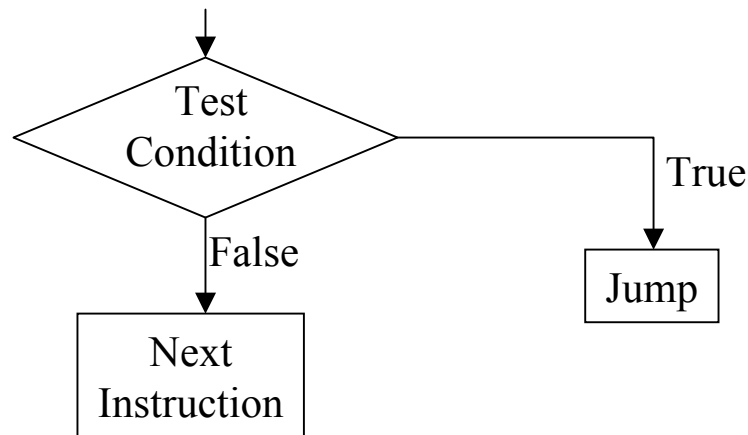
Program Branching Instructions

PROGRAM BRANCHING

ACALL	addr11	Absolute subroutine call	2	24
LCALL	addr16	Long subroutine call	3	24
RET		Return from subroutine	1	24
RETI		Return from interrupt	1	24
AJMP	addr11	Absolute jump	2	24
LJMP	addr16	Long jump	3	24
SJMP	rel	Short jump (relative addr)	2	24
JMP	@A+DPTR	Jump indirect relative to the DPTR	1	24
JZ	rel	Jump if Accumulator is zero	2	24
JNZ	rel	Jump if Accumulator is not zero	2	24
CJNE	A,direct,rel	Compare direct byte to A _{CC} and jump if not equal	3	24
CJNE	A,#data,rel	Compare immediate to A _{CC} and jump if not equal	3	24
CJNE	RN,#data,rel	Compare immediate to register and jump if not equal	3	24
CJNE	@Ri,#data,rel	Compare immediate to indirect and jump if not equal	3	24
DJNZ	Rn,rel	Decrement register and jump if not zero	2	24
DJNZ	direct,rel	Decrement direct byte and jump if not zero	3	24
NOP		No operation	1	12

Jump Instructions

- The 8051 has 2 types of JUMP instructions
- Unconditional Jump
 - This instruction type will load the PC with a new address and will automatically jump to the instruction at that address
- Conditional Jump
 - This instruction type will only jump if a certain condition is true
 - Similar to an “if” statement in C.



Unconditional Jumps

The 8051 has 3 unconditional jump instructions with a different range: -

- SJMP (Short Jump)
 - Allows a jump of -128 to $+127$ bytes relative to the current PC value
 - Instruction is 2 bytes long
- AJMP (Absolute Jump)
 - Allows a jump with the same 2KByte page that the PC is currently located in
 - Instruction is 2 bytes long
- LJMP (Long Jump)
 - Allows a jump anywhere within the 64KByte program memory range of the 8051
- If unsure which of the 3 instructions to use, simply use the JMP instruction and let the assembler decide which instruction to use.

Conditional Jumps

- The 8051 can test conditions at the bit and byte level
- Bit Conditional Jump Instructions
 - These instructions will jump if a bit is in a certain state
 - e.g. JC label ;jump to label if carry bit is set
 - JNC label2 ;jump to label2 if the carry bit is clear
 - These instructions are commonly used for arithmetic instructions and for the testing of flags
- Byte Conditional Jump Instructions
 - DJNZ – Decrement and Jump if Not Zero
 - CJNE – Compare and Jump if Not Equal